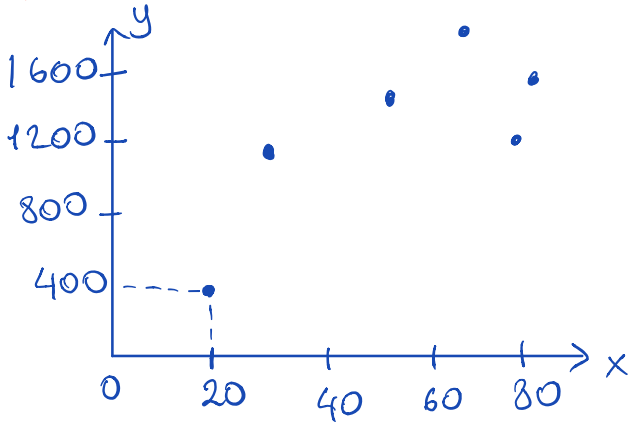
CHAPTER 14 – Linear Regression



Review of Statistics:

Goal: Summarize data using statistics that convey as much information as possible.

1) location of center of the distribution of data
2) degree of spread of the data set.

Measure of Location

$$\bar{y} = \frac{y_i}{n}$$ (arithmetic mean)

median: midpoint of a group of data (50th percentile)
- put the data in ascending order
- odd measurements ⇒ median is the middle value
- even " ⇒ arithmetic mean of the two middle values

7 3 4 9 5 → (order) 3 4 **5** 7 9 (median)

mode: value that occurs most frequently.

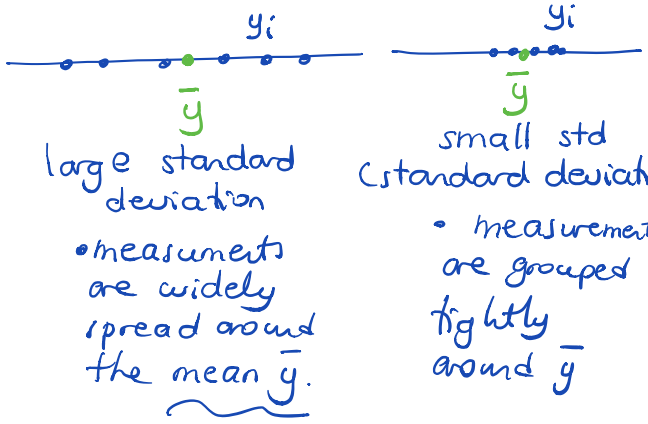
Measure of spread:

Range:
- Difference btwn. largest and smallest value
- Not reliable (sensitive to extreme values)

Standard deviation (or variance):

$$s_y^2 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2}{n-1}$$

$$= \frac{\sum_i y_i^2 - (\sum y_i^2)/n}{n-1}$$



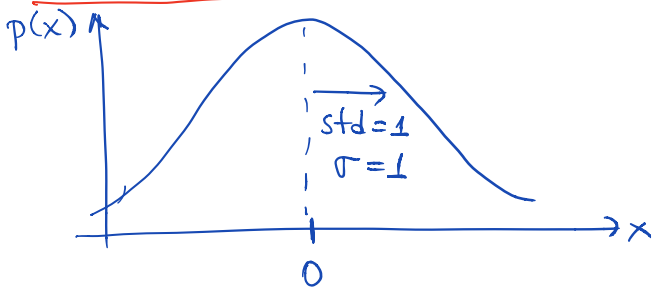
why $n-1$?

- $n-1$ degrees of freedom: if $\bar{y}$ is known and $y-1$ of the values are specified, the remaining value is fixed.
- Spread of a single data pt. is meaningless. When $n=1$, the result is meaningless (infinity)

coefficient of variation $\quad c.v. = \frac{s_y}{\bar{y}} \times 100\%$ (normalizes std. with mean)

The Normal Distribution

$x \sim N(0,1) = p(x)$



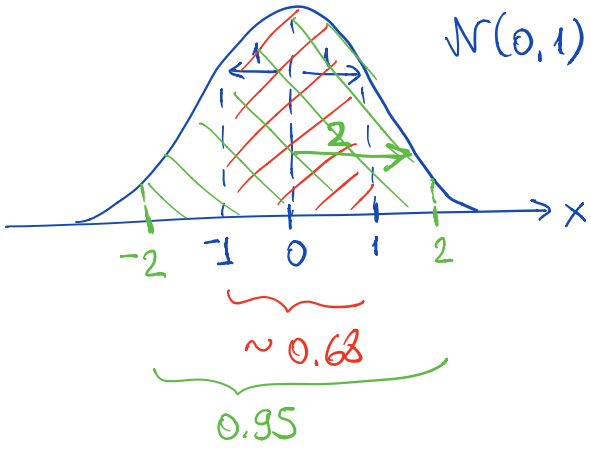
$N(\mu, \sigma)$ — μ: mean, σ: std.

Histogram: sort the measurements into intervals, bins.



$N(\bar{y}, s_y)$

- If we have a very large set of data, histogram can be approximated by a smooth curve (e.g. normal distribution)
- If a quantity is normally distributed, the range
  $(\bar{y}-s_y, \bar{y}+s_y)$ ⇝ contain ~68% of the total measurements
  $(\bar{y}-2s_y, \bar{y}+2s_y)$ ⇝ contain ~95% of measurements



* If $\bar{y} = 6.6$ and $s_y = 0.097133$ $\quad \bar{y}-2s_y \quad \bar{y}+2s_y$
  ⇒ 95% of readings in $(6.405735, 6.794266)$

MATLAB

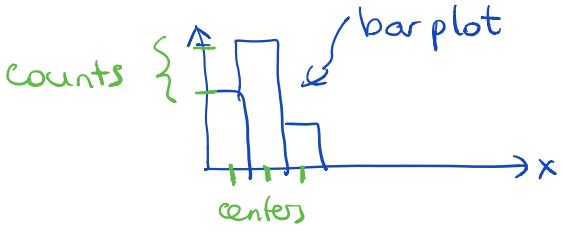
```
>> mean(s)
>> median(s)
>> mode(s)
>> min(s)
>> max(s)
>> var(s)
>> std(s)
>> [counts, centers] = hist(x, nbins)
```

counts, centers: output

x: vector of data

nbins: # bins

```
>> hist(s) % histogram bar plot with 10 bins.
```



Random numbers

```
>> r = rand(m,n) % random numbers uniformly distributed btwn. 0 and 1
```

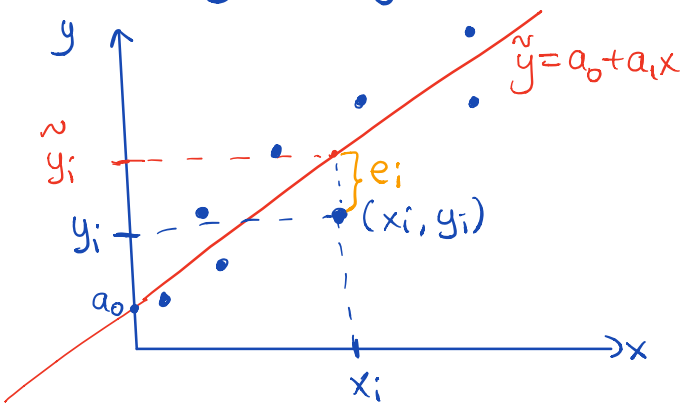
to generate uniform distribution on another interval: [low, up]

```
runiform = low + (up-low)*rand(m,n)
>> r = randn(m,n) % normally dist. mean=0, std=1
>>  rnormal = m + s*randn(m,n)
```

↳ normally dist mean = m, std = s

Linear Least-Squares Regression

* If there's error in the data, may fit a curve (e.g. line) that shows the general trend in the data, without necessarily passing thru the individual ptr.



Set of observations:

(x_1, y_1)

(x_2, y_2)

⋮

(x_n, y_n)

straight line

$$\tilde{y}_i = a_0 + a_1 x_i + e_i$$

$a_0 + a_1 x_i$: parameters of the line

e_i: residual between model and observations

$$e_i = y_i - a_0 - a_1 x_i$$

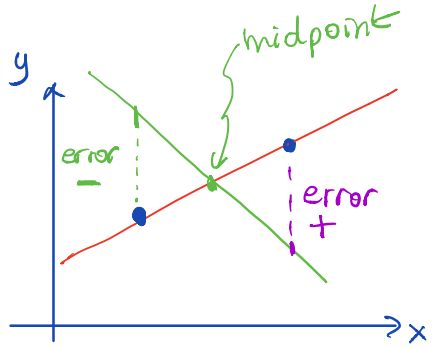
Criteria For Best Fit

$$\Rightarrow \sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)$$

↳ not suitable since:

→ Best line is the line connecting the 2 ptr (red line)

↳ But any line passing throughthe midpt. results in zero total error (min error)



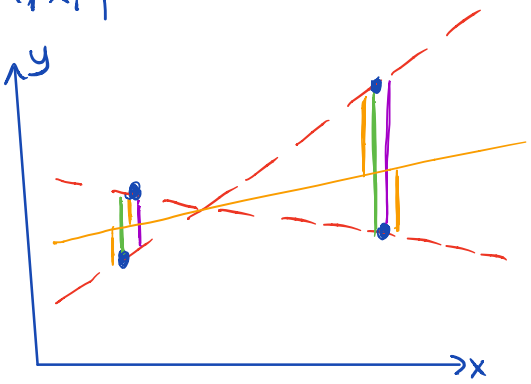
Data with 2 ptr.

$$\Rightarrow \sum_{i=1}^{n} |e_i| = \sum_{i=1}^{n} |y_i - a_0 - a_1 x_i|$$

↓

not suitable:

Any straight line (orange) falling btwn. the dashed red lines minimize sum of absolute values of residuals



Data with 4 pts.

Sum of Least squares of the residuals

$$S_r = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2 \quad \rightsquigarrow \text{least squares}$$

gives a unique line for a given set of data.

2 eq^{n}s
2 unknowns

$$\begin{cases} \dfrac{\partial S_r}{\partial a_0} = -2 \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i) = 0 \\ \dfrac{\partial S_r}{\partial a_1} = -2 \sum_{i=1}^{n} [(y_i - a_0 - a_1 x_i) x_i] = 0 \end{cases}$$

$$\begin{cases} \sum y_i - \sum a_0 - \sum_{i=1}^{n} a_1 x_i = 0 \\ \sum x_i y_i - \sum a_0 x_i - \sum a_1 x_i^2 = 0 \end{cases}$$

$$a_1 = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

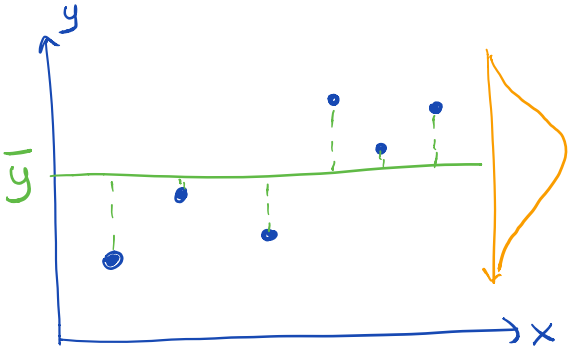
$\bar{y}$: mean of y; $\bar{x}$: mean of x

Quantification of Error of Linear Regression

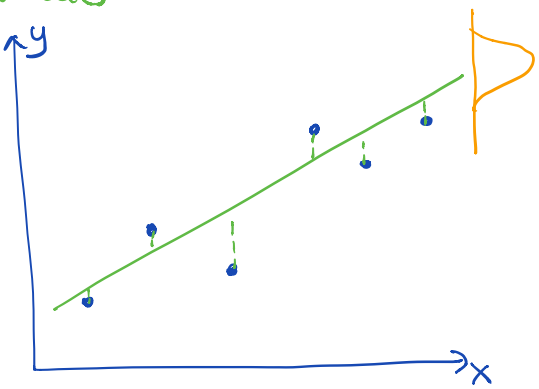
$$S_r = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2$$ Sum of squares of residual errors

$$S_t = \sum (y_i - \bar{y})^2$$

↳ mean of y values



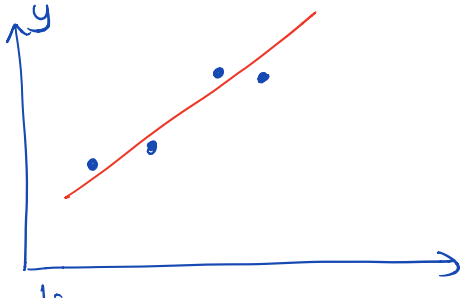
Spread of data around the mean $\bar{y}$



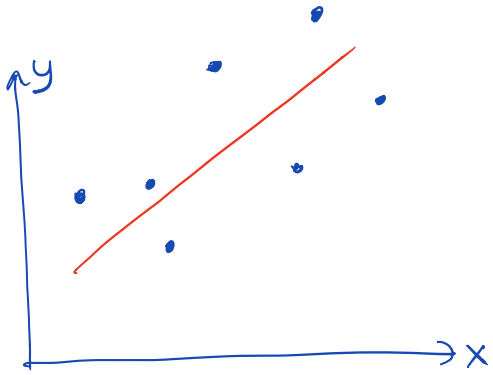
Spread of data around the best fit line

Standard error of estimate $= \sqrt{\frac{S_r}{n-2}}$

↓ like "standard deviation" for the regression line



linear regression with small residual errors



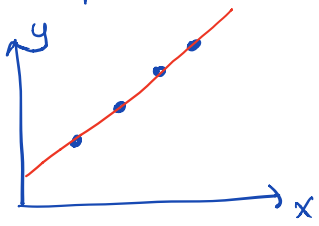
large residual errors

$S_t - S_r$ = improvement or error reduction due to using a straight line rather than $\bar{y}$.

$$r^2 = \frac{S_t - S_r}{S_t}$$, r: correlation coefficient

↳ coeff. of determination

perfect fit: $S_r = 0$ and $r^2 = 1$ (line explains 100% of the variability of the data)



$\Rightarrow r^2 = 0$, $S_r = S_t$ and the line fit gives no improvement

$$r = \frac{n\sum(x_i y_i) - (\sum x_i)(\sum y_i)}{\sqrt{n\sum x_i^2 - (\sum x_i)^2}\sqrt{n\sum y_i^2 - (\sum y_i)^2}}$$

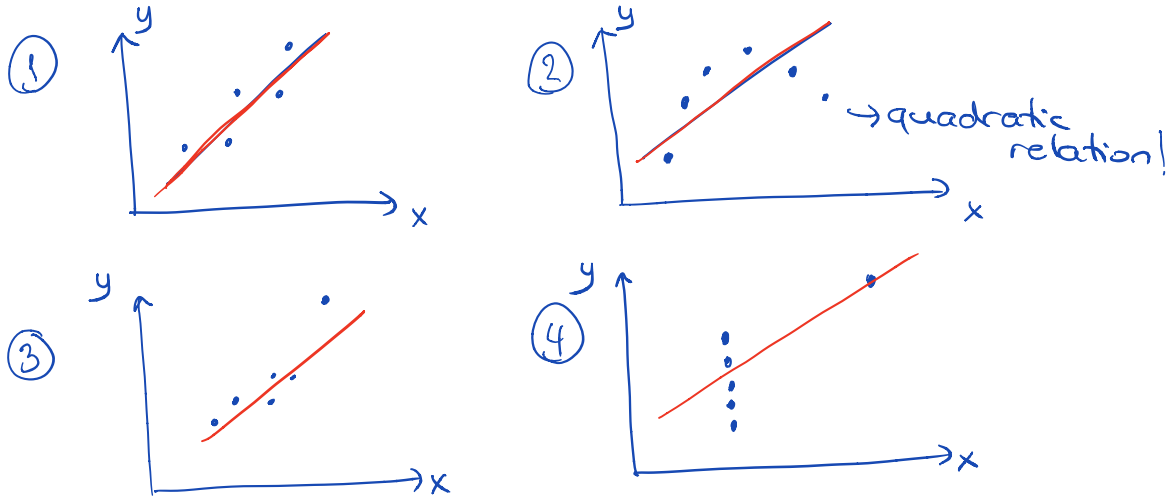
$0 \leq r \leq 1$

↳ x & y are linearly maximally correlated

↓ no linear correlation btwn x & y

$\Rightarrow$ r is close to 1, the line fit is "good".

$\Rightarrow$ it is possible to obtain high r, when the underlying relation btwn. x and y can be non-linear.



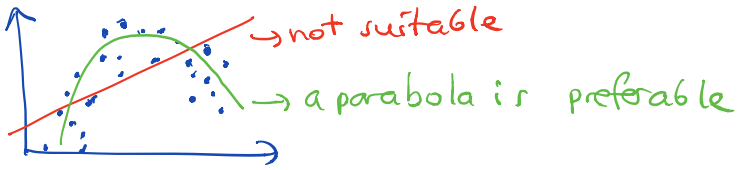
All 4 have the same best-fit line (red) : $y = 3 + 0.5x$

and $r^2 = 0.67$

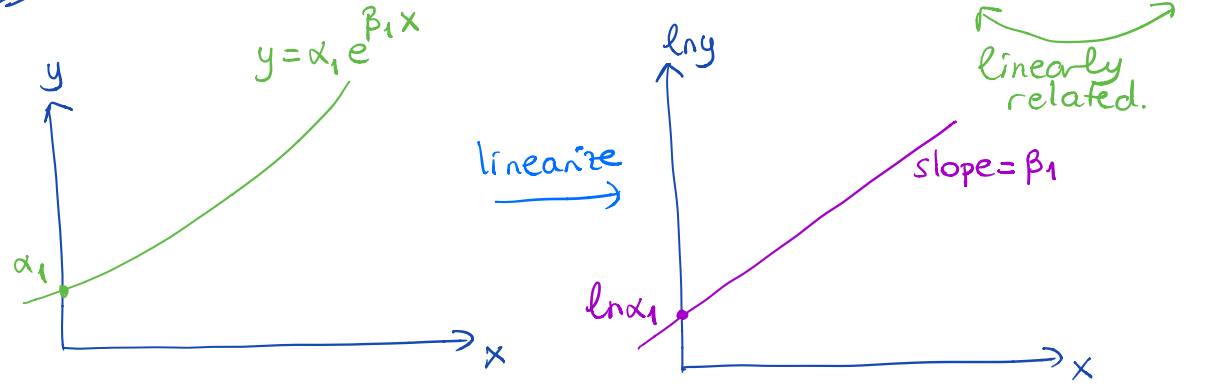
Linearization of Nonlinear Equations

⇒ Linear regression assumption : relation btwn x & y is linear.

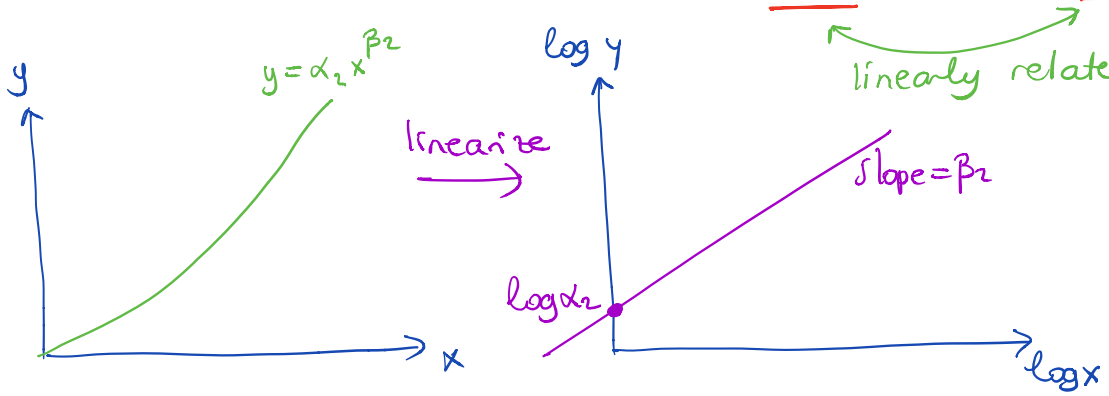
⇒ This not always the case.



EX Exponential model : $y = \alpha_1 e^{\beta_1 x} \xrightarrow{\ln} \ln y = \ln \alpha_1 + \beta_1 x$



⇒ power eq^n : $y = \alpha_2 x^{\beta_2}$ $\xrightarrow{\log}$ $\log y = \log \alpha_2 + \beta_2 \log x$



⇒ $\boxed{y = \alpha_3 \frac{x}{\beta_3 + x}}$ (used in population growth models)

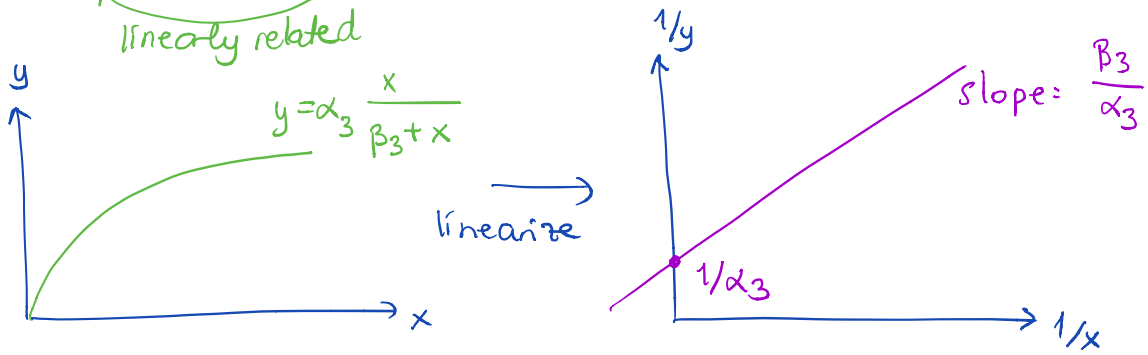
↓ Linearize

invert

$$\frac{1}{y} = \frac{1}{\alpha_3 \frac{x}{\beta_3 + x}} = \frac{\beta_3 + x}{\alpha_3 x} = \frac{1}{\alpha_3} + \frac{\beta_3}{\alpha_3} \frac{1}{x}$$

$$\frac{1}{y} = \frac{1}{\alpha_3} + \frac{\beta_3}{\alpha_3} \frac{1}{x}$$

linearly related



⇒ Nonlinear regression can be used.

⇒ Alternative : use linear regression after linearization.

Linearization of Nonlinear Equations

EXAMPLE 14.6 Fitting Data with the Power Equation

Problem Statement. Fit Eq. (14.23) to the data in Table 14.1 using a logarithmic transformation. $y = \alpha_2 x^{\beta_2}$ Linearize $\log y = \log \alpha_2 + \beta_2 \log x$

Solution. The data can be set up in tabular form and the necessary sums computed as in Table 14.6.

The means can be computed as

$$\bar{x} = \frac{12.606}{8} = 1.5757 \qquad \bar{y} = \frac{20.515}{8} = 2.5644$$

TABLE 14.6 Data and summations needed to fit the power model to the data from Table 14.1

i	x_i	y_i	$\log x_i$	$\log y_i$	$(\log x_i)^2$	$\log x_i \log y_i$
1	10	25	1.000	1.398	1.000	1.398
2	20	70	1.301	1.845	1.693	2.401
3	30	380	1.477	2.580	2.182	3.811
4	40	550	1.602	2.740	2.567	4.390
5	50	610	1.699	2.785	2.886	4.732
6	60	1220	1.778	3.086	3.162	5.488
7	70	830	1.845	2.919	3.404	5.386
8	80	1450	1.903	3.161	3.622	6.016
Σ			12.606	20.515	20.516	33.622

Linear Regression Equations are: Replace with $x_i, y_i \rightarrow \log x_i, \log y_i$; $x_i^2 \rightarrow (\log x_i)^2$

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2} \qquad a_0 = \bar{y} - a_1 \bar{x}$$

The slope and the intercept can then be calculated with Eqs. (14.15) and (14.16) as

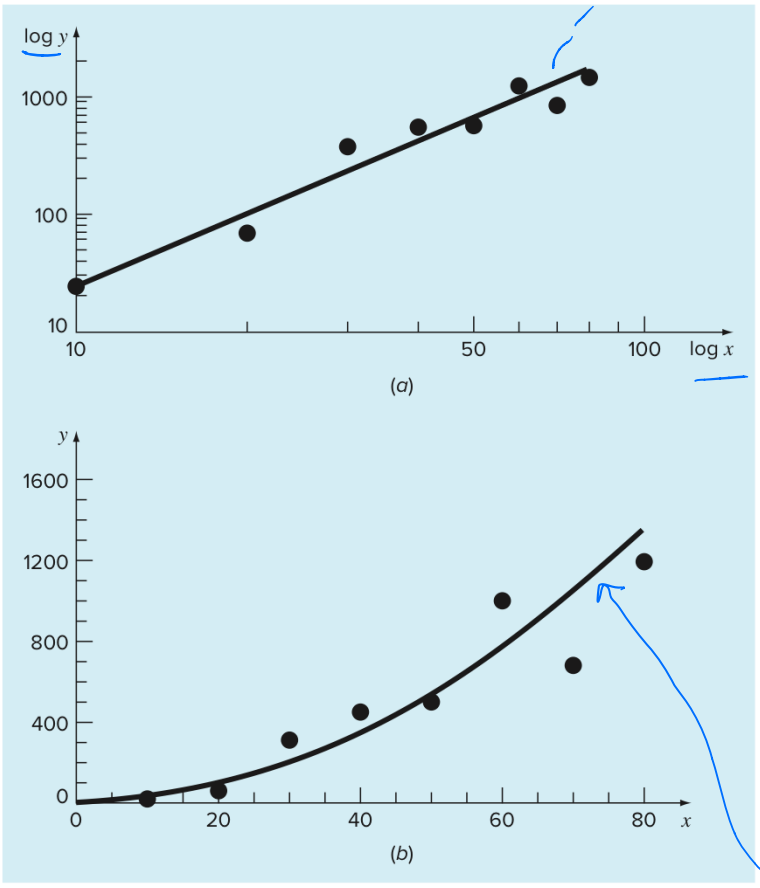
$$a_1 = \frac{8(33.622) - 12.606(20.515)}{8(20.516) - (12.606)^2} = 1.9842$$

$$a_0 = 2.5644 - 1.9842(1.5757) = -0.5620$$

The least-squares fit is

$$\log y = -0.5620 + 1.9842 \log x$$

The fit, along with the data, is shown in Fig. 14.14*a*.



$\log y = -0.5620 + 1.9842 \log x$

$= \log \alpha_2 + \beta_2 \log x$

$\log \alpha_2 = -0.5620$

$\Rightarrow \alpha_2 = 10^{-0.5620}$

$\alpha_2 = 0.2741$

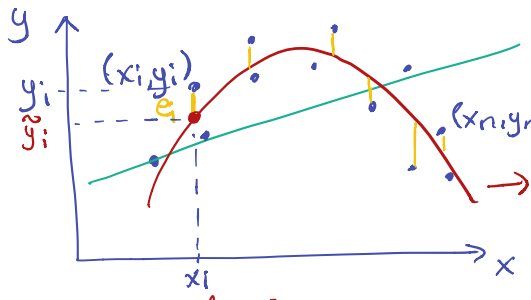
$\beta_2 = 1.9842$

$y = \alpha_2 x^{\beta_2}$

$y = (0.2741) x^{(1.9842)}$

FIGURE 14.14
Least-squares fit of a power model to the data from Table 14.1. (*a*) The fit of the transformed data. (*b*) The power equation fit along with the data.

CH 15 – General Linear Least-squares and Nonlinear Regression



Data is not suitable for linear regression

→ parabola is preferable

Polynomial Regression

e.g. Fit a 2nd order polynomial (or quadratic):

$$y = a_0 + a_1 x + a_2 x^2 + e$$

Sum of squares of the residuals:

$$S_r = \sum_{i=1}^{n} \underbrace{(y_i}_{\text{true}} - \underbrace{(a_0 + a_1 x_i + a_2 x_i^2)}_{\text{estimate}})^2 \quad \text{, unknowns: } a_0, a_1, a_2$$

e_i

To generate the least squares fit (minimize S_r)

$$\frac{\partial S_r}{\partial a_0} = 0 \Rightarrow -2 \sum_i (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

$$\frac{\partial S_r}{\partial a_1} = 0 \Rightarrow -2 \sum x_i (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

$$\frac{\partial S_r}{\partial a_2} = 0 \Rightarrow -2 \sum x_i^2 (y_i - a_0 - a_1 x_i - a_2 x_i^2) = 0$$

Arrange into normal equations:

$$\begin{aligned} n a_0 + (\sum x_i) a_1 + (\sum x_i^2) a_2 &= \sum y_i \\ (\sum x_i) a_0 + (\sum x_i^2) a_1 + (\sum x_i^3) a_2 &= \sum x_i y_i \\ (\sum x_i^2) a_0 + (\sum x_i^3) a_1 + (\sum x_i^4) a_2 &= \sum x_i^2 y_i \end{aligned}$$

3 unknowns, 3 eqns

unknowns

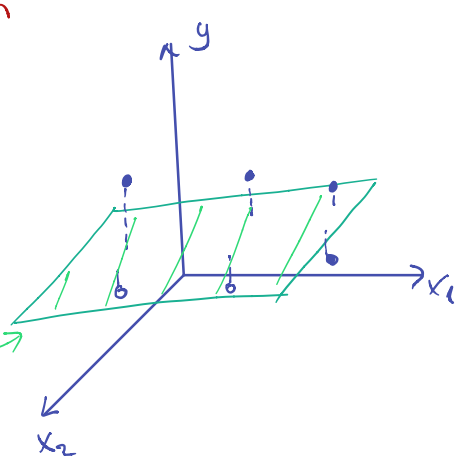
Extension to m^{th} order polynomial:

$$y = a_0 + a_1 x + a_2 x^2 + \cdots + a_m x^m + e$$

Multiple Linear Regression

$$y = a_0 + a_1 x_1 + a_2 x_2 + e$$

y is a linear function of two or more independent variables.



There's a regression plane

$$S_r = \sum_{i=1}^{n} (\underbrace{y_i}_{\text{true}} - \underbrace{(a_0 + a_1 x_{1,i} + a_2 x_{2,i})}_{\text{estimate}})^2$$

$$\frac{\partial S_r}{\partial a_0} = 0 \quad , \quad \frac{\partial S_r}{\partial a_1} = 0 \quad , \quad \frac{\partial S_r}{\partial a_2} = 0 \quad \left.\begin{matrix} \\ \\ \end{matrix}\right\} \begin{matrix} 3 \text{ eq}^n\text{s} \\ 3 \text{ unknowns} \end{matrix}$$

$$\begin{bmatrix} n & \sum x_{1,i} & \sum x_{2,i} \\ \sum x_{1,i} & \sum x_{1,i}^2 & \sum x_{1,i} x_{2,i} \\ \sum x_{2,i} & \sum x_{1,i} x_{2,i} & \sum x_{2,i}^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_{1,i} y_i \\ \sum x_{2,i} y_i \end{bmatrix}$$

Extension to m-dimension:

$$y = a_0 + a_1 x_1 + a_2 x_2 + \cdots + a_m x_m + e$$

Power Equations: $y = a_0 x_1^{a_1} x_2^{a_2} \cdots x_m^{a_m}$

↓ Linearize (Transformation)

$$\log y = \log a_0 + a_1 \log x_1 + \cdots + a_m \log x_m$$

Generalized Linear Least squares

Up to now:
- → linear regression
- → polynomial regression
- → multiple linear regression

General linear least squares model:

$$y = a_0 z_0 + a_1 z_1 + \cdots + a_m z_m + e \quad (1)$$

unknowns: $a_0, \ldots, a_m$

$z_0, \ldots, z_m$: basis function

Special case: Multiple Linear Regression

$$z_0 = 1, \; z_1 = x_1, \; z_2 = x_2 \; \cdots \; z_m = x_m$$

Special case: Polynomial Regression

$$z_0 = 1, \; z_1 = x, \; z_2 = x^2, \; \cdots \; z_m = x^m$$

Special case: $y = a_0 + a_1 \cos(\omega x) + a_2 \sin(\omega x)$ → Fourier analysis

"Linear" means: model's dependence on a's. Functions (basis) can be highly nonlinear.

Truly nonlinear model: $y = a_0 (1 - e^{-a_1 x})$ → can not be written in format (1)

$$y = a_0 z_0 + a_1 z_1 + a_2 z_2 + \cdots + a_m z_m + e \quad (1)$$

1st pt: $([z_{01}, z_{11}, z_{12} \cdots z_{m1}], y_1) \Rightarrow y_1 = a_0 z_{01} + a_1 z_{11} + \cdots + a_m z_{m1} + e_1$

Arrange eq^n (1) as: Data: (x_i, y_i), $i = 1, \ldots, n$

$$\underline{y} = \underline{\underline{z}}\,\underline{a} + \underline{e}$$

$$\underline{\underline{z}} = \begin{bmatrix} z_{01} & z_{11} & \cdots & z_{m1} \\ z_{02} & z_{12} & \cdots & z_{m2} \\ \vdots & & & \vdots \\ z_{0n} & z_{1n} & \cdots & z_{mn} \end{bmatrix} \quad (n) \times (m+1)$$

→ matrix of the calculated values of basis functions at the measured values of the independent variables.

m: # variables

n: # data points $\quad n \geq m+1$

$\underline{y}^T = [y_1 \; y_1 \cdots y_n]$, $\underline{a}^T = [a_0 \; a_1 \cdots a_m]$

$\underline{e}^T = [e_1 \; e_2 \cdots e_m]$

$$S_r = \sum_{i=1}^{n} \left(y_i - \sum_{j=0}^{m} a_j z_{ji} \right)^2$$

normal equations

$$\text{Set } \frac{\partial S_r}{\partial a_i} = 0 \Rightarrow \underline{y} = \underline{\underline{z}}\,\underline{a} \Rightarrow \underline{\underline{z}}^T \underline{y} = \underbrace{\underline{\underline{z}}^T \underline{\underline{z}}}_{(m+1)\times(m+1)}\,\underline{a}$$

invertible

not invertible ← $(n) \times (m+1)$

$$(\underline{\underline{z}}^T \underline{\underline{z}})^{-1} \underline{\underline{z}}^T \underline{y} = \underbrace{(\underline{\underline{z}}^T \underline{\underline{z}})^{-1} (\underline{\underline{z}}^T \underline{\underline{z}})}_{\underline{\underline{I}}}\, a$$

Least Squares Solution

$$\boxed{\underline{a} = (\underline{\underline{z}}^T \underline{\underline{z}})^{-1} \underline{\underline{z}}^T \underline{y}}$$

Nonlinear Regression

Fit a nonlinear model to data:

$y = a_0 (1 - e^{-a_1 x}) + e$ (can not linearize by taking ln of both sides)

unknowns: a_0, a_1

Least squares fit:

Data: (x_i, y_i) $i = 1, \ldots, n$

$$f(a_0, a_1) = \sum_{i=1}^{n} \left[\underbrace{y_i - a_0 (1 - e^{-a_1 x_i})}_{e_i} \right]^2$$

, solution must be iterative one.

MATLAB : fminsearch

$[x, fval] = fminsearch(fun, x_0, options, p1, p2, \ldots)$

↳ vector of initial gues

Linearization of Nonlinear Equations

EXAMPLE 14.6 Fitting Data with the Power Equation

Problem Statement. Fit Eq. (14.23) to the data in Table 14.1 using a logarithmic transformation. $y = \alpha_2 x^{\beta_2}$ Linearize $\log y = \log \alpha_2 + \beta_2 \log x$

Solution. The data can be set up in tabular form and the necessary sums computed as in Table 14.6.

The means can be computed as

$$\bar{x} = \frac{12.606}{8} = 1.5757 \qquad \bar{y} = \frac{20.515}{8} = 2.5644$$

TABLE 14.6 Data and summations needed to fit the power model to the data from Table 14.1

i	x_i	y_i	$\log x_i$	$\log y_i$	$(\log x_i)^2$	$\log x_i \log y_i$
1	10	25	1.000	1.398	1.000	1.398
2	20	70	1.301	1.845	1.693	2.401
3	30	380	1.477	2.580	2.182	3.811
4	40	550	1.602	2.740	2.567	4.390
5	50	610	1.699	2.785	2.886	4.732
6	60	1220	1.778	3.086	3.162	5.488
7	70	830	1.845	2.919	3.404	5.386
8	80	1450	1.903	3.161	3.622	6.016
Σ			12.606	20.515	20.516	33.622

Linear Regression Equations are: Replace with $x_i, y_i \rightarrow \log x_i, \log y_i$; $x_i^2 \rightarrow (\log x_i)^2$

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2} \qquad a_0 = \bar{y} - a_1 \bar{x}$$

The slope and the intercept can then be calculated with Eqs. (14.15) and (14.16) as

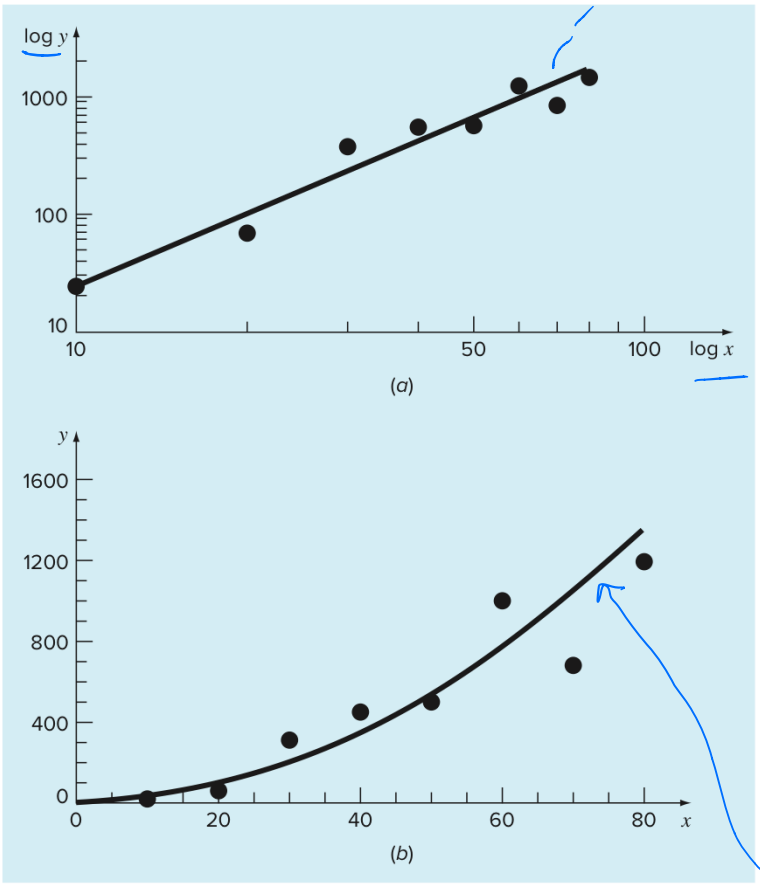
$$a_1 = \frac{8(33.622) - 12.606(20.515)}{8(20.516) - (12.606)^2} = 1.9842$$

$$a_0 = 2.5644 - 1.9842(1.5757) = -0.5620$$

The least-squares fit is

$$\log y = -0.5620 + 1.9842 \log x$$

The fit, along with the data, is shown in Fig. 14.14*a*.



$\log y = -0.5620 + 1.9842 \log x$

$= \log \alpha_2 + \beta_2 \log x$

$\log \alpha_2 = -0.5620$

$\Rightarrow \alpha_2 = 10^{-0.5620}$

$\alpha_2 = 0.2741$

$\beta_2 = 1.9842$

$y = \alpha_2 x^{\beta_2}$

$y = (0.2741) x^{(1.9842)}$

FIGURE 14.14
Least-squares fit of a power model to the data from Table 14.1. (*a*) The fit of the transformed data. (*b*) The power equation fit along with the data.

USING MATLAB

① Write a function to compute the sum of squares:

$$S_r = \sum_{m=1}^{n} \left(y_m - a_1 x_m^{a_2}\right)^2$$

y_m: true; $a_1 x_m^{a_2}$: estimated

```
function f = fSSR(a,xm,ym)
    yp = a(1) * xm.^a2;    % predicted
    f = sum((ym-yp).^2);
```

(a = [a1 a2]ᵀ) } save in fSSR.m

② Enter Data

```
>> x = [10,20,30,40,50,60,70,80];
>> y = [25,70,380,550,610,1220,830,1450];
>> fminsearch(@fSSR,[1,1],[],x,y);
```

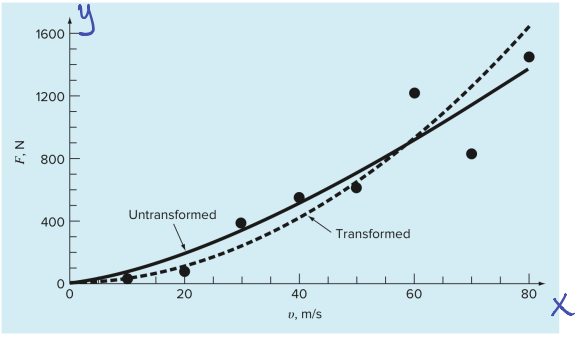
([1,1]: initial guess)

```
>> ans =
    2.5384    1.4359
```

$y = 2.5384\, x^{1.4359}$ (untransformed)

$\underbrace{y = (0.2741)\, x^{(1.9842)}}_{\text{transformed}}$

FIGURE 15.4
Comparison of transformed and untransformed model fits for force versus velocity data f Table 14.1.



Although the fits are different, it is difficult to judge which one is better.

Polynomial Fitting in MATLAB

polyfit: fits a least-squares n^{th} order polynomial.

```
>> p = polyfit(x,y,n)
```

- n → order of polynomial
- y → dependent variable
- x → independent variable

$$f(x) = p_1 x^n + p_2 x^{n-1} + \cdots + p_n x + p_{n+1}$$

```
>> p = polyfit(x,y,1)
```
↳ returns a line (slope & intercept)

```
>> x = [10,20,30,40,50,60,70,80];
>> y = [25,70,380,550,610,1220,830,1450];
>> a = polyfit(x,y,1);
```

$a = 19.47 \quad -234.28$

$\Rightarrow \boxed{y = 19.47x - 234.28.}$

polyval: compute a value using coefficients

```
>> y = polyval(p,x)
```

- p → polynomial coefficients
- y → the best-fit value at x.